

Workflow Management System

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Guide:

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Introduction

Workflow Management System is designed to streamline tasks in organization by providing a structured way to assign, monitor, and complete tasks. This system enables managers to create workflows, allocate tasks, and track employee progress while ensuring effective communication.



Problem Statement

Most existing workflow management platforms are complex and paid , and not user-friendly, making them difficult for small teams and new users to adapt to. The complex interfaces and steep learning curves lead to inefficiencies, reducing productivity and making task management more time-consuming. Additionally, the lack of seamless collaboration tools affects communication and coordination, making it harder for teams to work efficiently.



Purpose and Need

- **Purpose:**

- Streamline task management
- Enhance collaboration
- Improve productivity
- Employees can update progress

- **Need:**

Address challenges in organizations like:

- Lack of transparency
- Time-consuming manual processes
- Communication gaps
- Difficulty tracking employee performance

Workflow Management System helps to overcome these challenges.



Objectives

- Design a user-friendly workflow management system
- Simplify task management by removing unnecessary complexity
- Focus on essential features for small to medium-sized teams
- Enhance collaboration through:
 - Seamless task tracking
 - Real-time communication through meetings

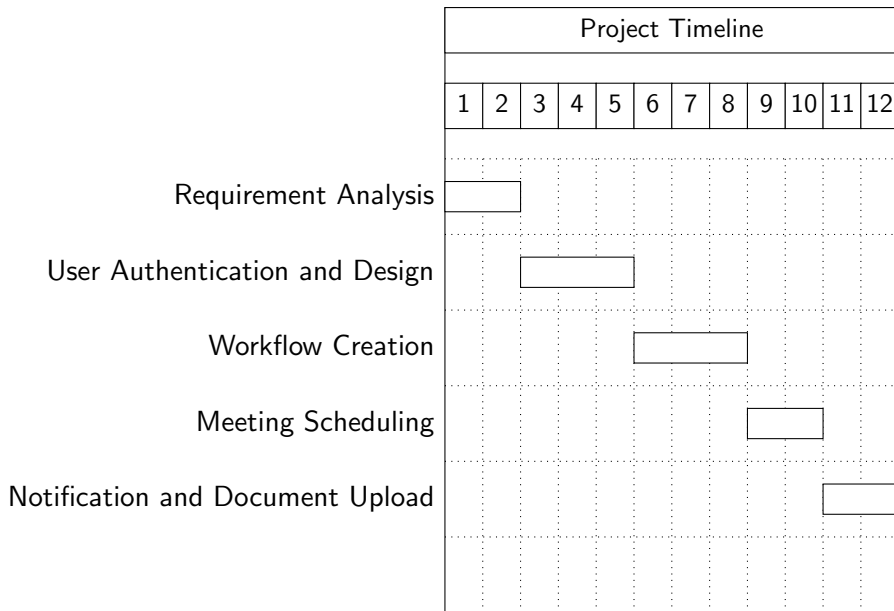


- Designed for small to medium-sized teams
- Offers essential task management features
- User-friendly interface
- Integrated video conferencing
- Role-based access control for secure authentication

- Many existing platforms are complex and difficult to navigate, making them less accessible for smaller teams.
- This project aims to bridge that gap by providing a simplified yet efficient system that integrates task management with real-time communication.
- By enhancing coordination, minimizing miscommunication, and streamlining task execution, it creates a structured and organized work environment.



Project Scheduling



Functional Requirements:

• User Management:

- Requirement ID: FR-AUTH-001
- Description: Implement user registration functionality using secure authentication methods such as username/password
- Input: User credentials
- Output: Authenticated session with appropriate access tokens.

• Role based access:

- Requirement ID: FR-AUTH-002
- Description: Validate user access permissions.
- Input: Credentials.
- Output: Validation status.



● **Dashboard Module:**

- Requirement ID: FR-AUTH-003
- Description: Visualize key metrics, task statuses, and system performance with customizable reports.
- Input: Task and user activity data.
- Output: Visual dashboard and analytical reports.

● **Workflow Creation:**

- Requirement ID: FR-AUTH-004
- Description: Uses visual elements like charts, notifications. Provides real-time updates and analytical reports.
- Input: Workflow creation data and user input.
- Output: Generated workflow with real-time updates and reports.



- **Task management:**

- Requirement ID: FR-AUTH-005
- Description: Create, assign, and manage tasks with features like setting deadlines and priorities.
- Input: Task details (title, description, priority, deadline, assignee)
- Output: tasks are created, assigned and can be displayed.

- **Document Upload :**

- Requirement ID: FR-AUTH-006

- :Description: Upload, store, and manage task-related documents with version control, ensuring role-based access.
- Input: documents can be selected or drag and dropped to upload.
- Output: uploaded documents can be seen and downloaded.



Software Requirement Specification

● **Communication:**

- Requirement ID: FR-AUTH-007
- Description: The manager can schedule meetings on google meet with timings and add employees
- Input: meeting details (title, start time, end time, select employees)
- Output: meetings are created and the employees get to join the meeting from their dashboard.

● **Notifications :**

- Requirement ID: FR-AUTH-008
- Description: system shall send email to the employees when tasks are assigned to them with task details and deadlines.
- Input: Task creation
- Output: emails are send to the employees when tasks are assigned to them.



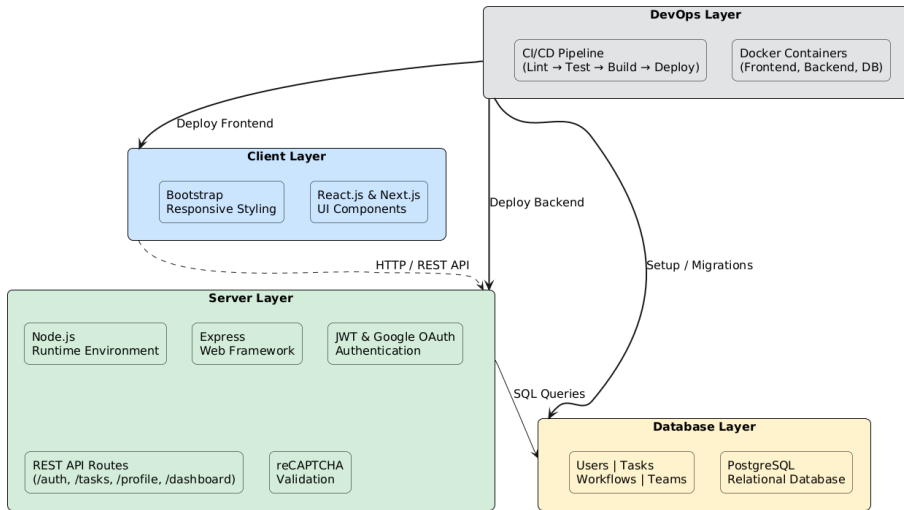
Non-Functional Requirements:

- **Security:** User authentication must be handled using JWT and Google OAuth for secure access.reCAPTCHA must be implemented for login and registration to prevent bot attacks.
- Efficient UI design for ease of use.
- **Secure API Data Transmission:** All API communications must be secured using HTTPS, ensuring encrypted data transfer between the React frontend and the PostgreSQL backend.
- **Access Control Role Management:** The system must implement role-based access control (RBAC), restricting access to sensitive features based on user roles (e.g., Manager, Team Member).

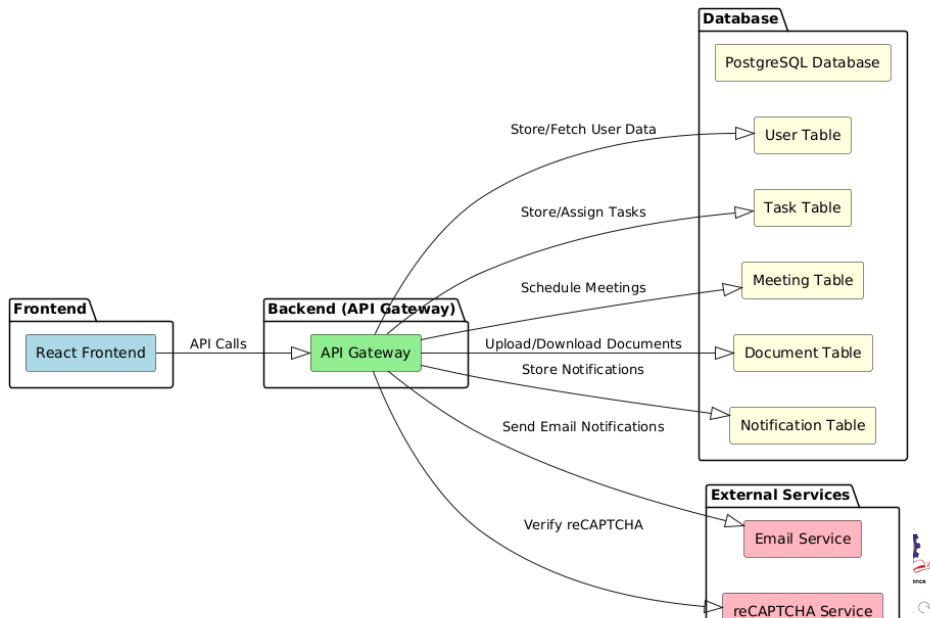


System Architecture

Workflow Management System Architecture

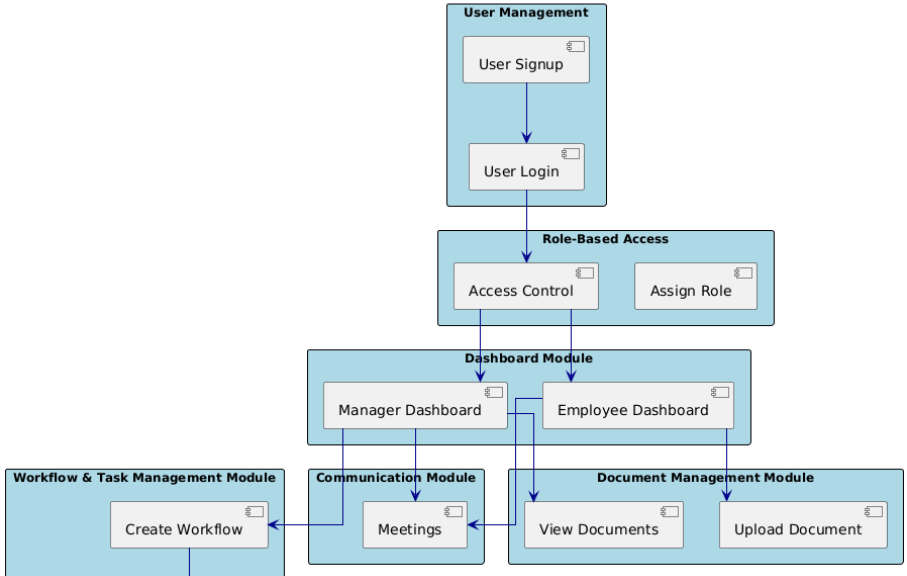


Application Architecture



Module Diagram

Module Diagram



Technology Stack

Frontend: ReactJS , Bootstrap

Backend: NodeJs ,ExpressJs

Database: PostgreSQL

Authentication: bcrypt is used for secure password and authentication. JSON Web Tokens for user authentication and authorization.

Communications and meeting: Google meet

Development Environment: VS Code

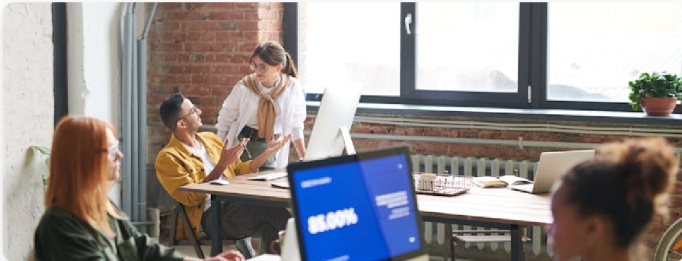
CI/CD: Gitlab ci/cd.framework :Jest





Transform Your Team's Productivity!

Streamline workflows, track progress, and collaborate seamlessly with your team using our intuitive management system designed for the modern workplace.





Task Management

Create, assign and track tasks with ease.



Workflow Automation

Automate repetitive processes and save time.



Performance Tracking

Monitor team performance with detailed analytics.

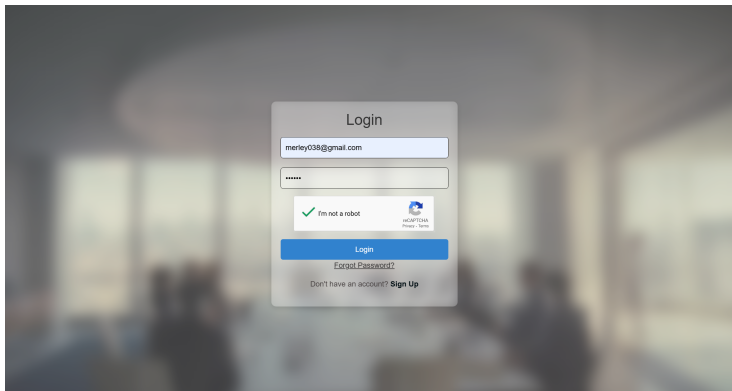


Meeting Scheduler

Schedule and manage meetings efficiently.

[Get Started](#)

[Create Account](#)



Sign Up

Employee

☐

I'm not a robot


hCAPTCHA
Privacy - Terms

Sign Up

Already have an account? [Login](#)

 Profile

Workflows

Tasks

Employees

LM Reports

[Schedule Meeting](#)

LM My Meetings

[Sign Out](#)

Create Workflow

Workflow Name

test

Description

demo

Add Tasks

Task Name

Assign to Employee

Medium

17-04-2025

Add Task

Tasks Added:

task1 - merley038@gmail.com (Medium)

Remove

[Create Workflow](#)

Scheduled Meetings

demo2

3/14/2025, 2:32:00 PM - 3/14/2025, 4:32:00 PM

Participants: John Doe, Jane Smith

4

[Join Meeting](#)

demo2

3/14/2025, 2:32:00 PM - 3/14/2025, 4:32:00 PM

Participants: John Doe, Jane Smith

6

[Join Meeting](#)

demo

3/1/2025, 1:16:00 PM - 3/19/2025, 1:17:00 PM

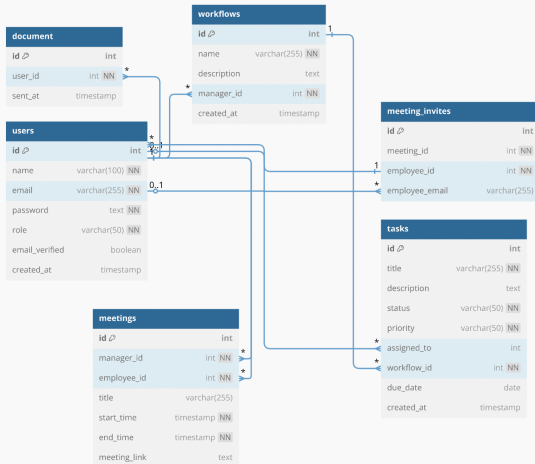
Participants: None

6

[Join Meeting](#)

demo

Database design



POST/api/auth/login - User login,generate verification link

POST/api/auth/register - User Registration

POST/api/workflows/create - Workflow creation

PUT/api/update-employee-profile - Update Profile

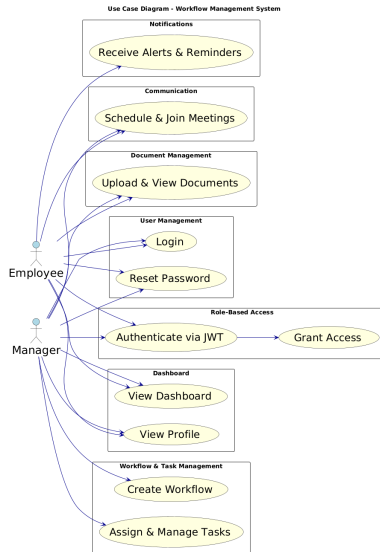
DELETE/api/users - Delete users

GET/api/users - List registered users

GET/api/tasks/assigned - List assigned tasks for employees

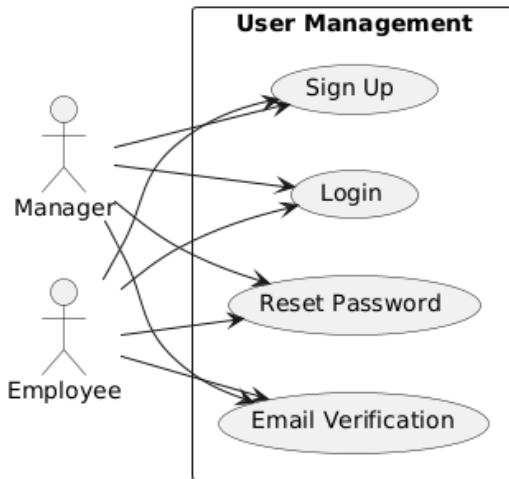


Usecase diagram

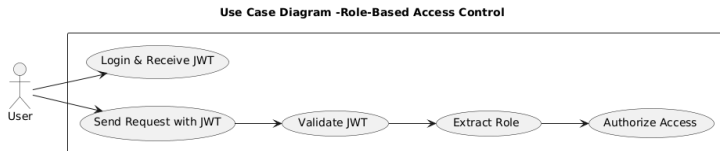


User management module

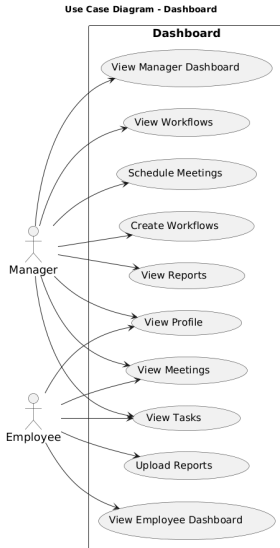
Use Case Diagram - User Management



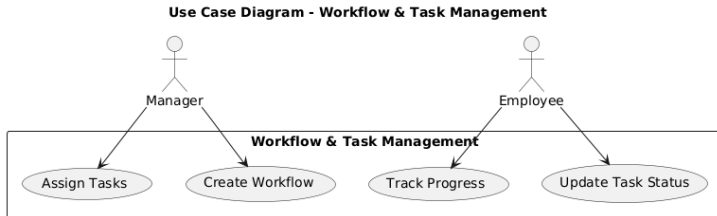
Role based access



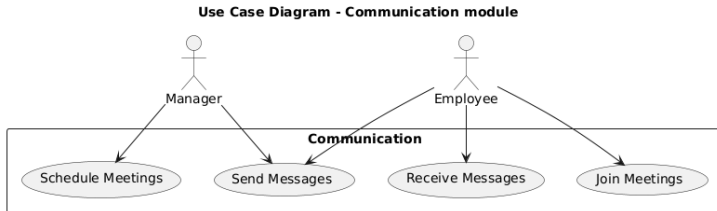
Dashboard module



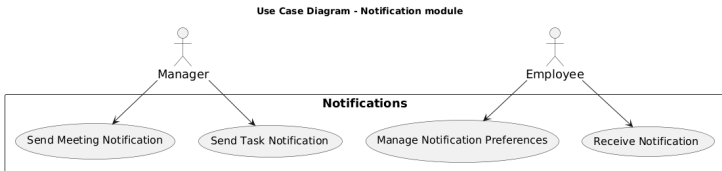
Workflow creation & Task management module



Communication module

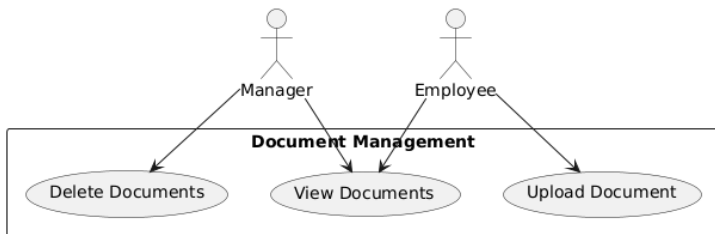


Notification module



Document upload module

Use Case Diagram - Document Management



User Authentication Role-Based Access

- ① User Signup/Login:
 - Input: username, email, password
 - Hash password and store securely
 - On login, validate credentials and issue token
- ② Role-Based Access Control:
 - Fetch user role (Manager or Employee)
 - Redirect to the appropriate dashboard



Dashboard Handling

- **Manager:** View task summary, meetings, documents, and employee progress
- **Employee:** View assigned tasks, meetings, and document uploads



Workflow

Task Management

- 1 Create Workflow (Manager assigns steps and members)
- 2 Assign Tasks (Define title, description, deadline, priority)
- 3 Employee updates status (Pending → In Progress → Completed)
- 4 Notify manager on task updates



Communication Notification System

- 1 Meetings (Video conference for team discussions)
- 2 Task Notifications (Updates, reminders, alerts)



Document Management

- ① Employees upload reports and files
- ② Managers review and approve/reject documents



Workflow management system follows best practices for clean and maintainable code:

- Follows consistent naming conventions:
 - camelCase for variables and functions.
 - PascalCase for classes and React components.
- Implements error handling with proper status codes and messages.
- Uses environment variables for sensitive configurations.



Setup Environment

Prerequisites: Install Node.js, **PostgreSQL**, and a package manager **npm**

Cloning the Repository: `git clone`

`https://gitlab.com/csd24-25/workflow-management-system/workflow-management.git`

Installing Dependencies:

`npm install`

Environment Configuration:

Create a `.env` file and set up environment variables.

Starting the Application:

`npm start`



Unit Testing Details

FRONT END

Function	Test Case	Expected Result	Actual Result
Sign up	Try to register using an already registered email id	Show "email already exist" error	Show "email already exist" error
Sign Up	Enter different passwords in password and confirm password fields	Show "passwords do not match" error	Show "passwords do not match" error
Login	Enter valid email and password	Successful login	Successful login



Unit Testing Details

BACK END

API Endpoint	Test Case	Expected Result	Actual Result
POST /api/auth/login	Valid credentials	Return JWT and Success	Return JWT and Success
POST /api/tasks/create	Missing required fields	Return error response	Return error response
GET /api/workflows	Fetch all workflows	Return workflow list	Return workflow list



Bug Reports and Resolution

Reportee	Bug Descriptions	Resolutions
Shahina	- User unable to log in - Sidebar buttons not navigating to correct route	- Fixed incorrect API routing - Fixed React Router paths
Josiya	- Email verification not sent - Role-based redirection not happening after login	- Fixed nodemailer config - Fixed role check in login success handler
Merlin	- Workflow not saving correctly - Task assignment mail not received	- Debugged SQL insert query - Fixed email loop logic and added error logs
Meghna	- Workflow not appearing in manager dashboard -Sidebar buttons not navigating to correct route	- Corrected database query -Fixed React Router paths

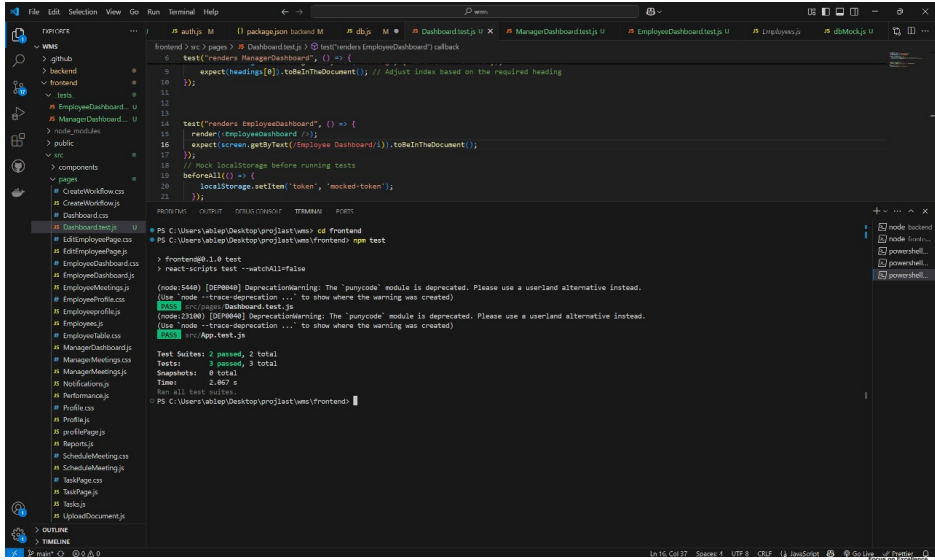


Status of Work & Contribution

- Josiya Jose: Document upload module, employee dashboard, login verification.
- Meghna B: Recaptcha verification, workflow and task creation module, notifications.
- Merlin Pynadath: Manager dashboard, meeting scheduling frontend, created workflows display.
- Shahina Abdul Salam: Meeting scheduling and display, CI/CD, Docker.



Supporting Visuals



Supporting Visuals

The screenshot displays the Docker Desktop interface. At the top, the 'docker.desktop' title bar is visible with a 'PERSONAL' label and a search bar. Below the title bar, a navigation sidebar on the left contains icons for containers, images, volumes, and networks. The main area shows a project named 'workflow-management' located at 'C:\Users\shah\OneDrive\Desktop\clone1\workflow management'. A 'View configurations' button is present in the top right of the project view.

Three containers are listed under the project:

- workflow-managem...** (postgres:15) with ID 5432-5432
- workflow-managem...** (workflow-managemer) with ID 5000-5000
- workflow-managem...** (workflow-managemer) with ID 3000-3000

The terminal output for the 'workflow-managemer' container shows the following:

```
Compiled with warnings.
2025-04-05 12:18:01 frontend-1 |
2025-04-05 12:18:01 frontend-1 | [eslint]
2025-04-05 12:18:01 frontend-1 | src/pages/EmployeeDashboard.js
2025-04-05 12:18:01 frontend-1 | Line 3:49: 'User' is defined but never used no-unused-vars
2025-04-05 12:18:01 frontend-1 | Line 9:25: 'setNotifications' is assigned a value but never used no-unused-vars
2025-04-05 12:18:01 frontend-1 | Search for the keywords to learn more about each warning.
2025-04-05 12:18:01 frontend-1 | To ignore, add // eslint-disable-next-line to the line before.
2025-04-05 12:18:01 frontend-1 |
2025-04-05 12:18:01 frontend-1 | WARNING in [eslint]
2025-04-05 12:18:01 frontend-1 | src/pages/EmployeeDashboard.js
2025-04-05 12:18:01 frontend-1 | Line 3:49: 'User' is defined but never used no-unused-vars
2025-04-05 12:18:01 frontend-1 | Line 9:25: 'setNotifications' is assigned a value but never used no-unused-vars
2025-04-05 12:18:01 frontend-1 |
2025-04-05 12:18:01 frontend-1 | webpack compiled with 1 warning
2025-04-05 12:18:36 backend-1 | reCAPTCHA response from Google: {
2025-04-05 12:18:36 backend-1 |   success: true,
2025-04-05 12:18:36 backend-1 |   challenge_ts: '2025-04-05T06:46:25Z',
2025-04-05 12:18:36 backend-1 |   hostname: 'localhost'
2025-04-05 12:18:36 backend-1 | }
2025-04-05 12:19:12 backend-1 | reCAPTCHA response from Google: {
2025-04-05 12:19:12 backend-1 |   success: true,
2025-04-05 12:19:12 backend-1 |   challenge_ts: '2025-04-05T06:46:48Z',
2025-04-05 12:19:12 backend-1 |   hostname: 'localhost'
2025-04-05 12:19:12 backend-1 | }
2025-04-05 12:19:16 backend-1 | Email sent successfully to: studentfisat@gmail.com
2025-04-05 12:19:45 backend-1 | reCAPTCHA response from Google: {
2025-04-05 12:19:45 backend-1 |   success: true,
```

The bottom status bar indicates system resources: RAM 3.45 GB, CPU 0.42%, and Disk 12.36 GB used (limit 1006.85 GB). The Windows taskbar at the very bottom shows the date as April 5, 2025, and the time as 05:11.

Conclusion

The Workflow Management System enhances task organization, team collaboration, and overall efficiency. It offers a structured approach to task assignment, progress tracking, and workflow automation. Key features such as email verification, reCAPTCHA integration, and role-based access control strengthen system security and user validation. This project demonstrates a practical and scalable solution for managing tasks in a collaborative environment.

